AN IMMERSION IS CLOSED IFF ITS IMAGE IS CLOSED

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 2071-2099, following the statement at 2065-2069.

Allowed standard prerequisites (editorial).

- Closed immersions as closed subspace structure. lemma-closed-immersion (same chapter, cited at ~2096). Identifies Y -> U with the closed subspace associated to I.
- Closed subspace associated to a sheaf of ideals. example-closed-subspace (same chapter, cited at ~2092). Canonical isomorphism of the closed subspaces of I and J.
- Glueing sheaves from data. Sheaves, Section sheaves-section-glueing-sheaves (cross-chapter, cited at ~2082). Glues I and O_{X \ i(Y)} to the sheaf of ideals J on X.

Source reference. Original reference 01HM: sources/stacks-schemes/pinned-original-source.tex, source lines 586–586. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 01HO: sources/stacks-schemes/pinned-original-source.tex, source lines 617–617. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00AK: focused-seeds/geometry/S9-schemes-immersion-when-closed/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $f : Y \to X$ be an immersion of schemes. Then f is a closed immersion if and only if $f(Y) \subset X$ is a closed subset.*

Proof. If f is a closed immersion then $f(Y)$ is closed by definition. Conversely, suppose that $f(Y)$ is closed. By definition there exists an open subscheme $U \subset X$ such that f is the composition of a closed immersion $i : Y \to U$ and the open immersion $j : U \to X$. Let $\mathcal{I} \subset \mathcal{O}_U$ be the quasi-coherent sheaf of ideals associated to the closed immersion i. Note that $\mathcal{I}|_{U \setminus i(Y)} = \mathcal{O}_{U \setminus i(Y)} = \mathcal{O}_{X \setminus i(Y)}|_{U \setminus i(Y)}$. Thus we may glue (see Sheaves, Section 00AK) $\mathcal{I}$ and $\mathcal{O}_{X \setminus i(Y)}$ to a sheaf of ideals $\mathcal{J} \subset \mathcal{O}_X$. Since every point of X has a neighbourhood where $\mathcal{J}$ is quasi-coherent, we see that

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$\mathcal{J}$ is quasi-coherent (in particular locally generated by sections). By construction $\mathcal{O}_X/\mathcal{J}$ is supported on U and, restricted there, equal to $\mathcal{O}_U/\mathcal{I}$. Thus we see that the closed subspaces associated to $\mathcal{I}$ and $\mathcal{J}$ are canonically isomorphic, see Example 01HM. In particular the closed subspace of U associated to $\mathcal{I}$ is isomorphic to a closed subspace of X. Since $Y \to U$ is identified with the closed subspace associated to $\mathcal{I}$, see Lemma 01HO, we conclude that $Y \to U \to X$ is a closed immersion. $\square$